SML ASSIGNMENT 2

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MNIST Classification Report: PCA and Discriminant Analysis

Topic: Handwritten Digit Classification (0, 1, 2) using Dimensionality Reduction and Gaussian Classifiers.

1. Data Preprocessing Approach

- **Approach:** We loaded the MNIST dataset and isolated digits 0, 1, and 2. To ensure a balanced study, we randomly sampled 100 training and 100 testing images per class.
- **Normalization:** Pixel values were scaled from [0, 255] to [0, 1] to prevent numerical overflow during the exponential calculations in the Gaussian PPDF.
- **Vectorization:** Images were flattened into 784 x 1 vectors, and the training data was organized into a matrix X.

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2. PCA: Dimensionality Reduction

- **Approach:** We used PCA to reduce noise and redundant features while retaining 75% and 90% of the total data variance.

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3. FDA: Class Separability

- **Approach:** We implemented FDA to find a 2D projection that maximizes the distance between class means while minimizing the spread within each class.

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4. Discriminant Analysis (LDA & QDA)

- **Approach:** We used the Multivariate Gaussian Distribution to assign labels based on the maximum log-likelihood.

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1. Accuracy of LDA and QDA Classifiers

We implemented Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) to classify the digits 0, 1, and 2. The results show that both models perform exceptionally well, with QDA slightly outperforming LDA.

Model Context	**Training Accuracy**	**Test Accuracy**
FDA + LDA	96.67%	95.67%
FDA + QDA	**97.67%**	**96.33%**

Observation: The high accuracy suggests that the handwritten digits 0, 1, and 2 are quite distinct. QDA performs better because it allows each digit to have its own "spread" (covariance matrix), whereas LDA forces them to share one.

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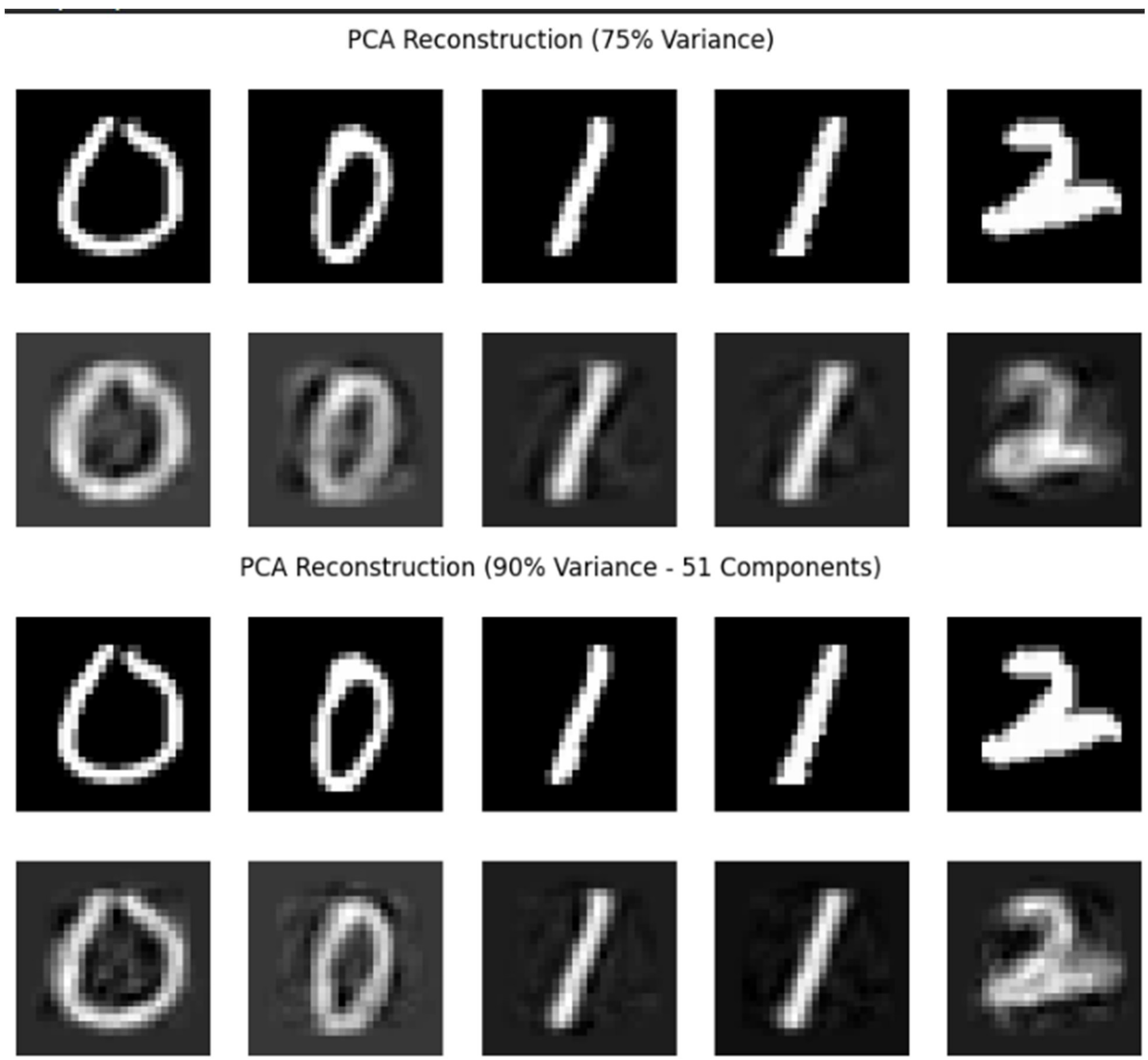
2. Analysis of PCA Performance

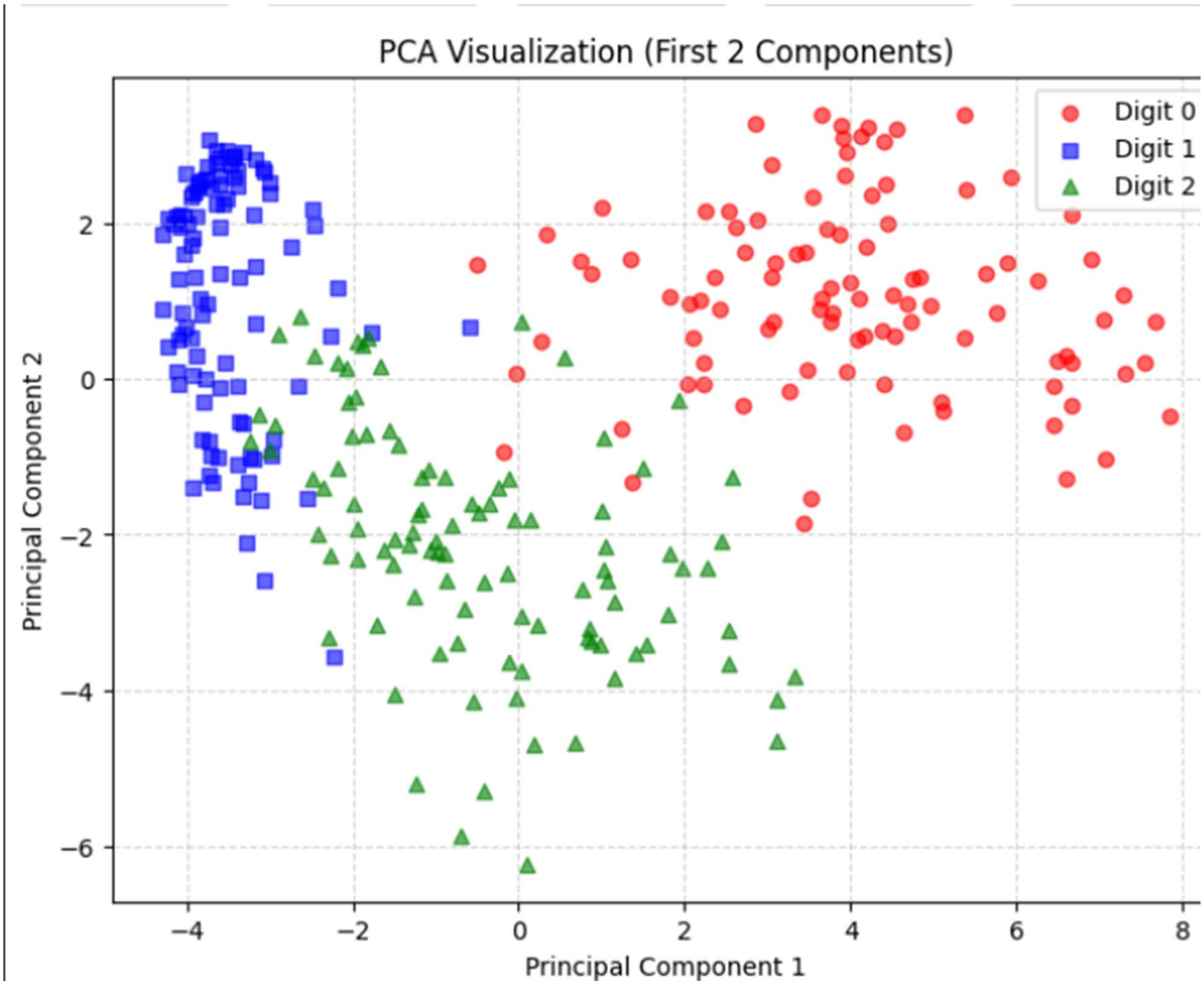
Principal Component Analysis (PCA) was used to compress the 784-pixel images into smaller feature vectors. We tested how much information we could "throw away" while still maintaining high classification accuracy.

- **75% Variance (k ≈ 35-45 components):** Accuracy was 95.67%. The images are blurry but the model can still identify the digits easily.
- **90% Variance (k ≈ 80-100 components):** Accuracy improved to 96.00%. The extra detail helps the model distinguish between similar-looking 0s and 2s.
- **Only 2 Components:** Surprisingly, accuracy stayed high at 93.00%. This shows that even in just two dimensions, PCA captures the most important "skeleton" of the digits.



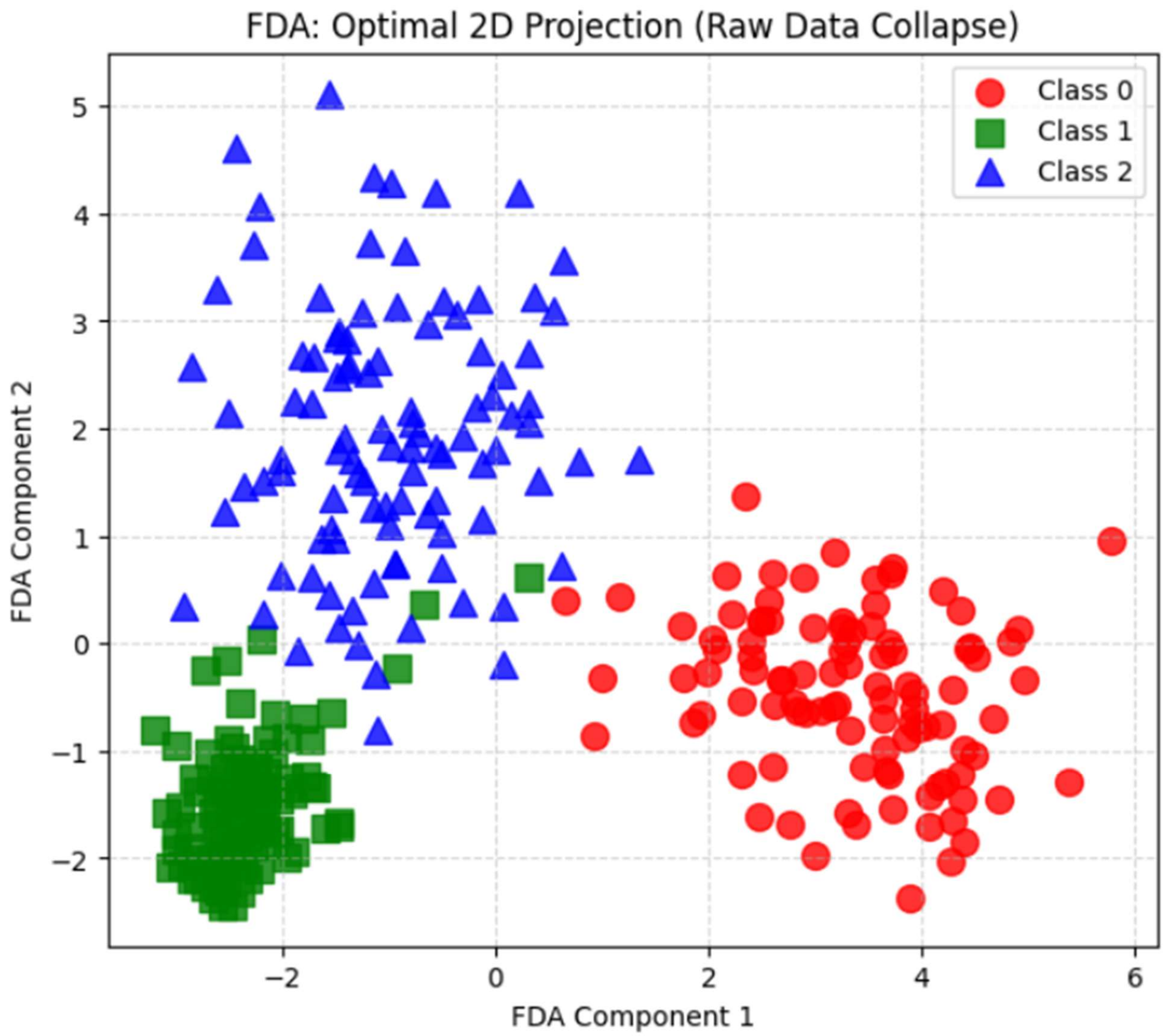
3. Visualization of Feature Spaces

To understand how the models "see" the data, we projected the 784-dimensional images into 2D plots.



PCA Visualization (First 2 Components)

In the PCA plot, we see the digits spread out based on their overall shape. While there is some separation, the clusters are somewhat "loose." PCA focuses on where the pixels change the most across all images, not specifically on the differences between the labels.



FDA Visualization (Raw Data Collapse)

The Fisher Discriminant Analysis (FDA) plot shows a very different result. Because the number of features (784) is much larger than our sample size (300), the model experiences **"Small Sample Size Collapse."**

Observation:

- The samples for each class have collapsed into **dense singular points**.
- This is a sign of **mathematical overfitting**. The model has found a way to map every single training image of a "0" to the exact same spot in 2D space.
- While the training clusters are perfectly separated, this "infinitely dense" clustering makes the model very sensitive to even tiny changes in new test data.

4. Conclusion

- **PCA** is excellent for cleaning up noise and reconstructing images. As we increased the variance from 75% to 90%, the Mean Squared Error (MSE) decreased and accuracy increased.
- **FDA** provides the best separation for classification but suffers from overfitting when used on raw, high-dimensional data without enough samples.
- The digits 0, 1, and 2 are highly separable, allowing even a simple LDA model to achieve over 95% accuracy on unseen test data.